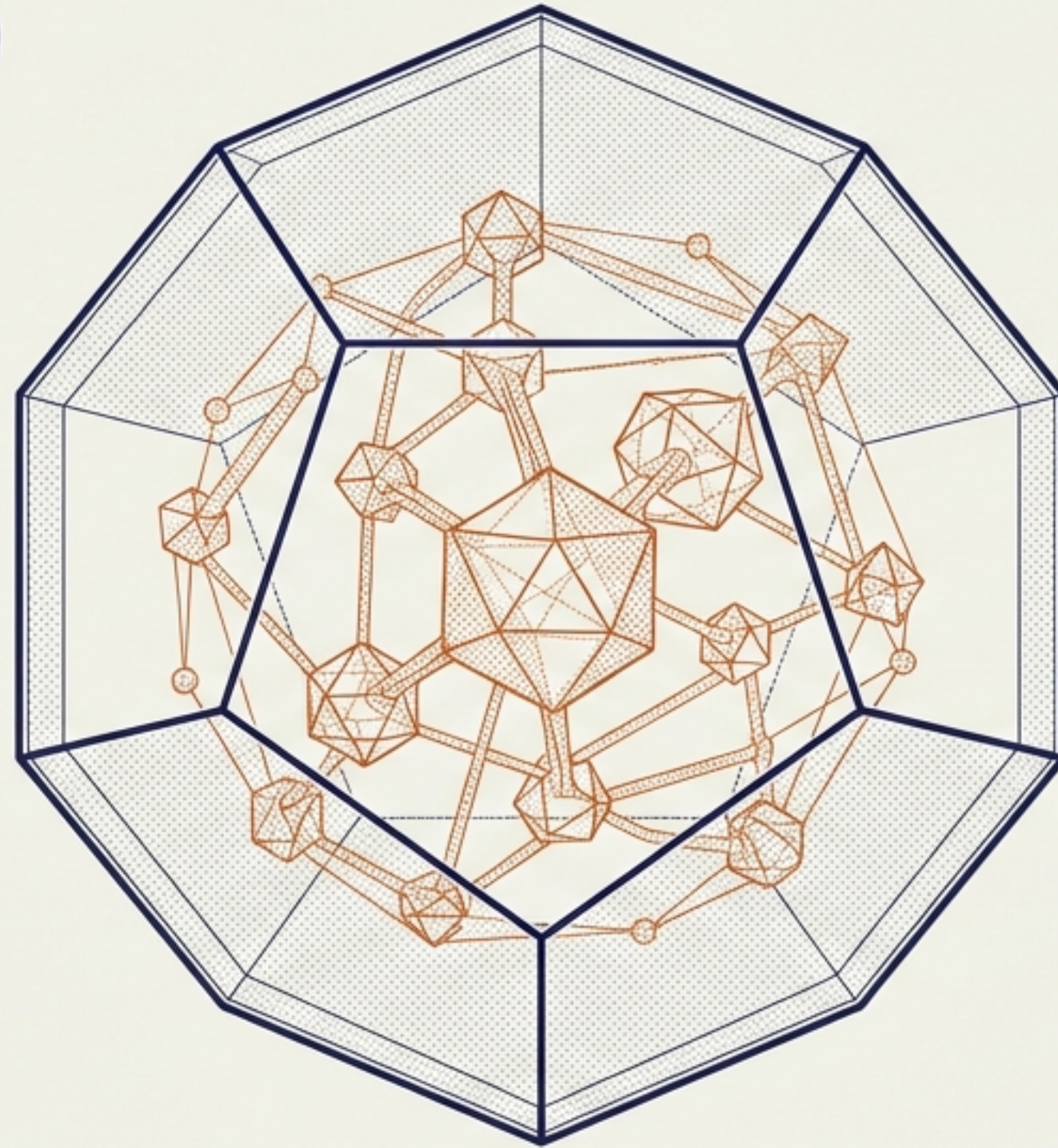
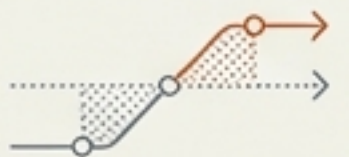


Graphs of Graphs of Graphs (G3)

Capturing Transformations,
Provenance, and Context in
Modern Knowledge Systems

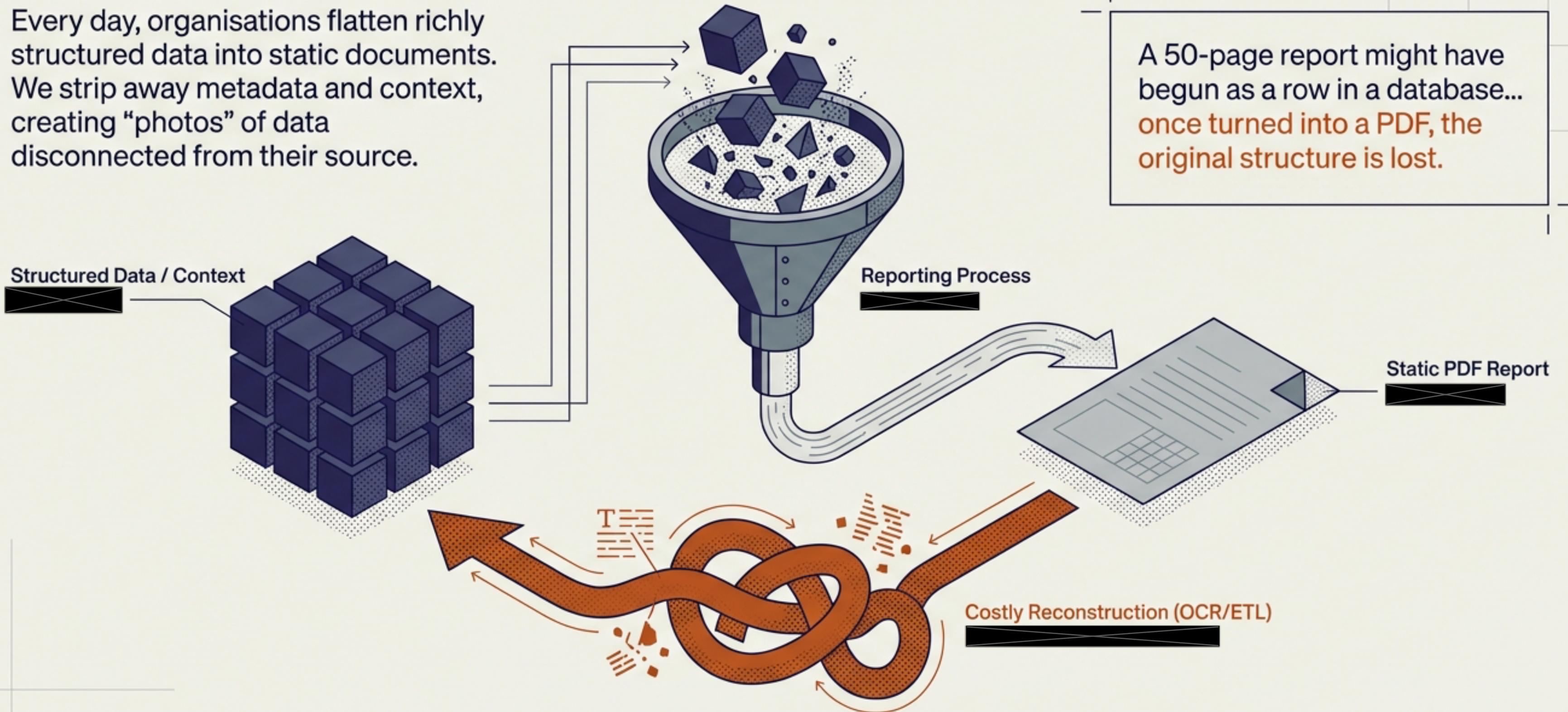


From static snapshots to living lineages.



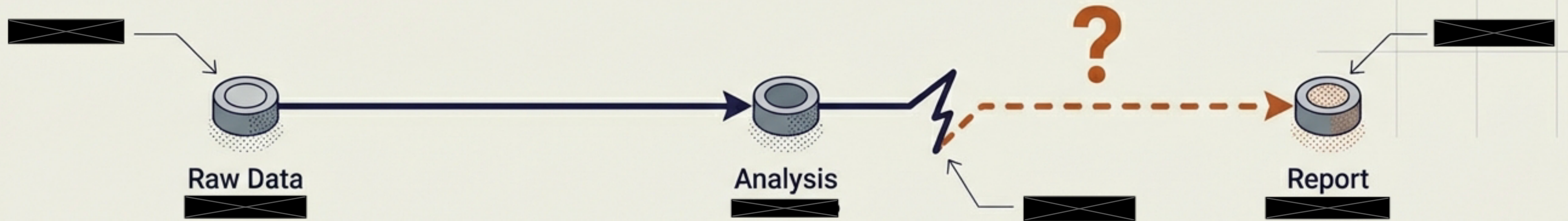
The Entropy Problem: We Are “Unbaking the Cake”

Every day, organisations flatten richly structured data into static documents. We strip away metadata and context, creating “photos” of data disconnected from their source.



The High Cost of Lost Lineage

Static artifacts tell us *what* happened, but not *why* or *how*.



Broken Trust

Without a decision trace, audits are arduous. We cannot verify if a KPI was derived from accurate data or a spreadsheet error.

AI Hallucinations

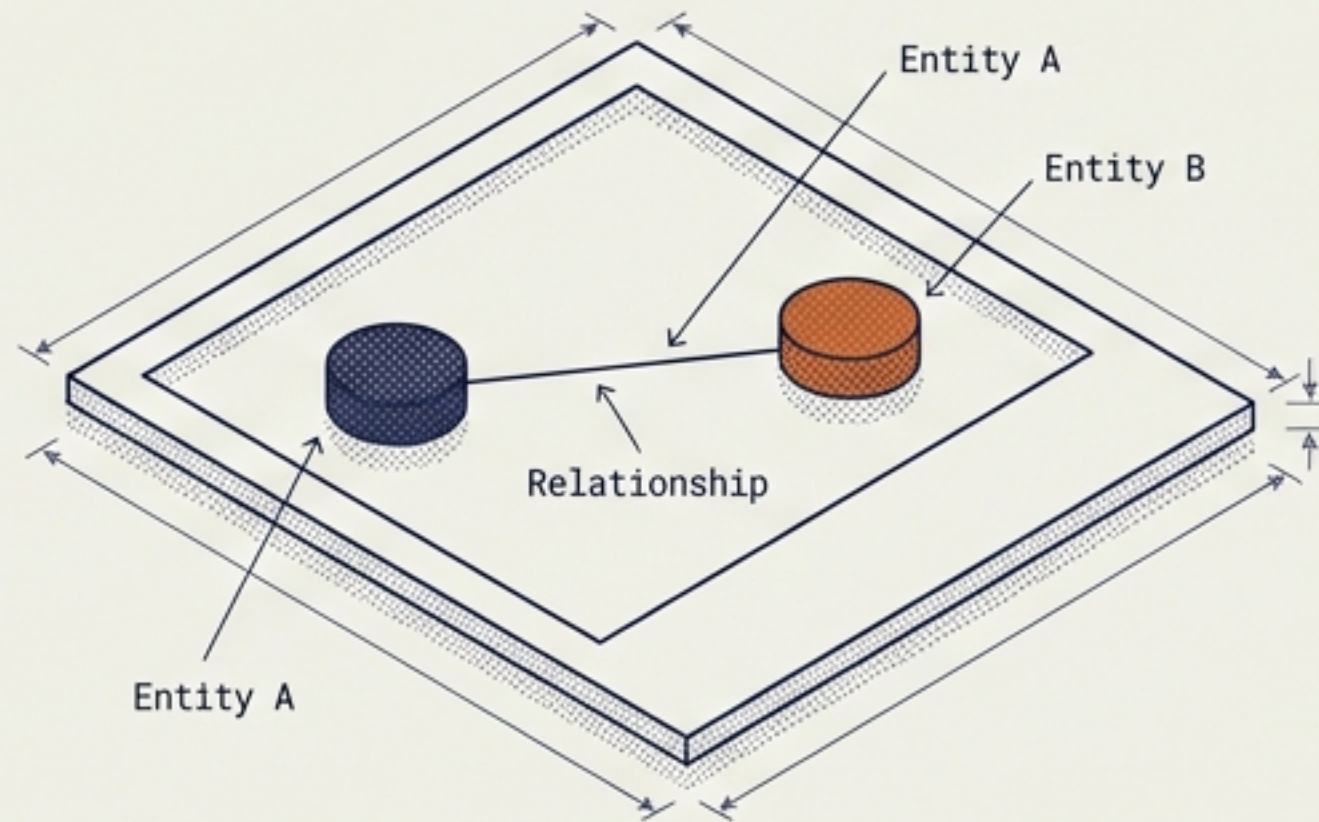
Models trained on flat text lack grounding in causality. They digest the claim without the evidence.

The 'Snapshot' Trap

There is no such thing as naturally unstructured data. It is an artificial artifact of our documentation process.

Evolution: From Snapshots to Cinema

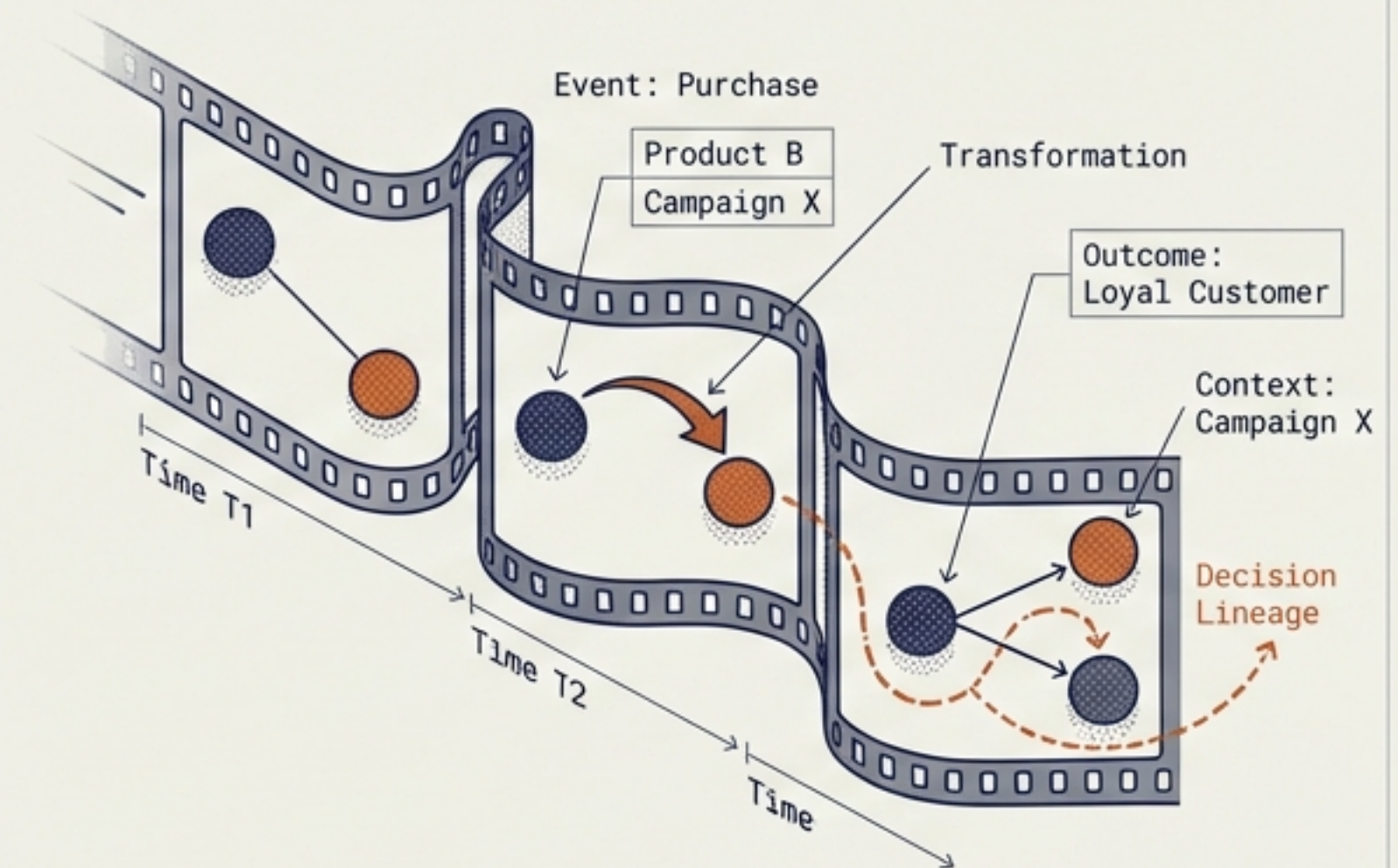
State-Oriented (Nouns)



Focus on Entities and static relationships.
Good for facts, bad for history.

Customer A purchased Product B.

Action-Oriented (Verbs)

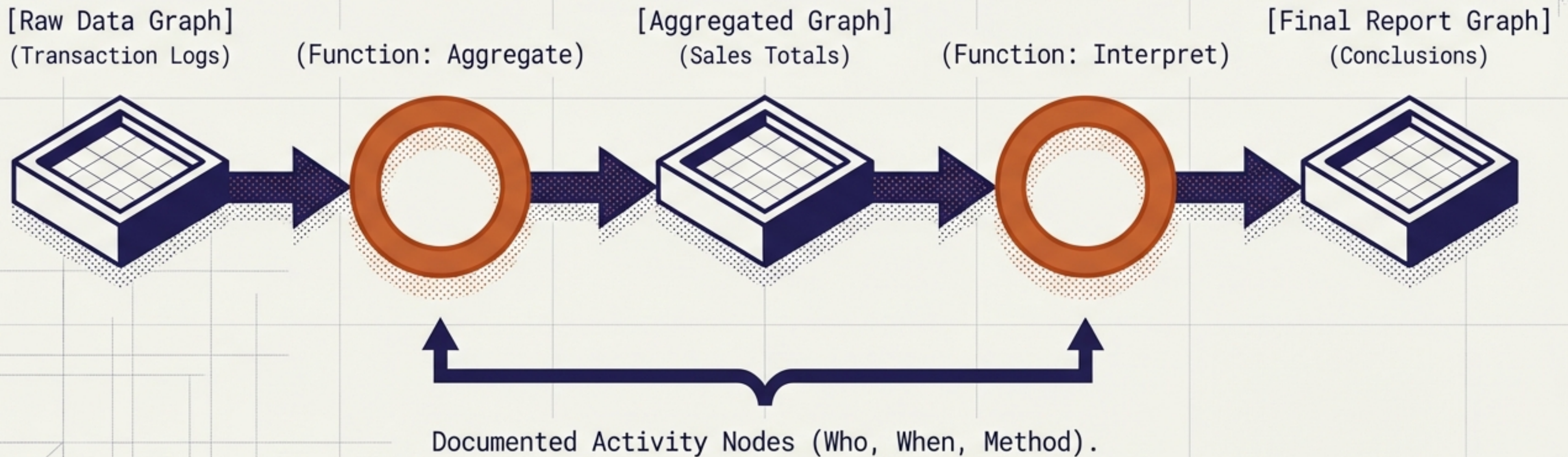


Focus on Events, Transformations, and Time.
Knowledge Graph + Time + Decision Lineage.

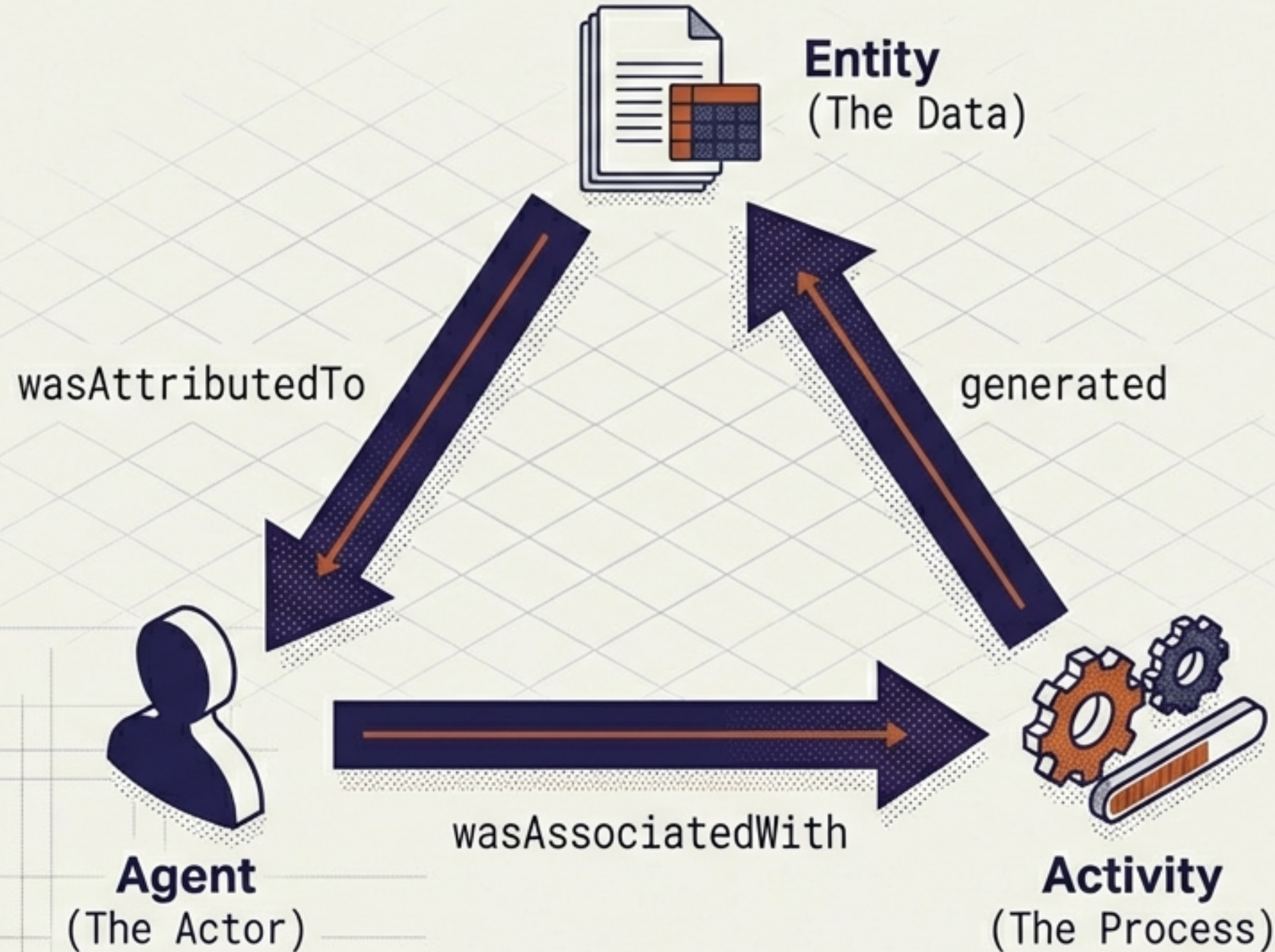
Customer A bought Product B *because of* Campaign X on Date Y.

The Core Mechanism: Capturing Transformations

Every analytical step is a function that transforms data. To preserve context, we must record the function, not just the result.



Standards-Based Provenance (W3C PROV)



Context:

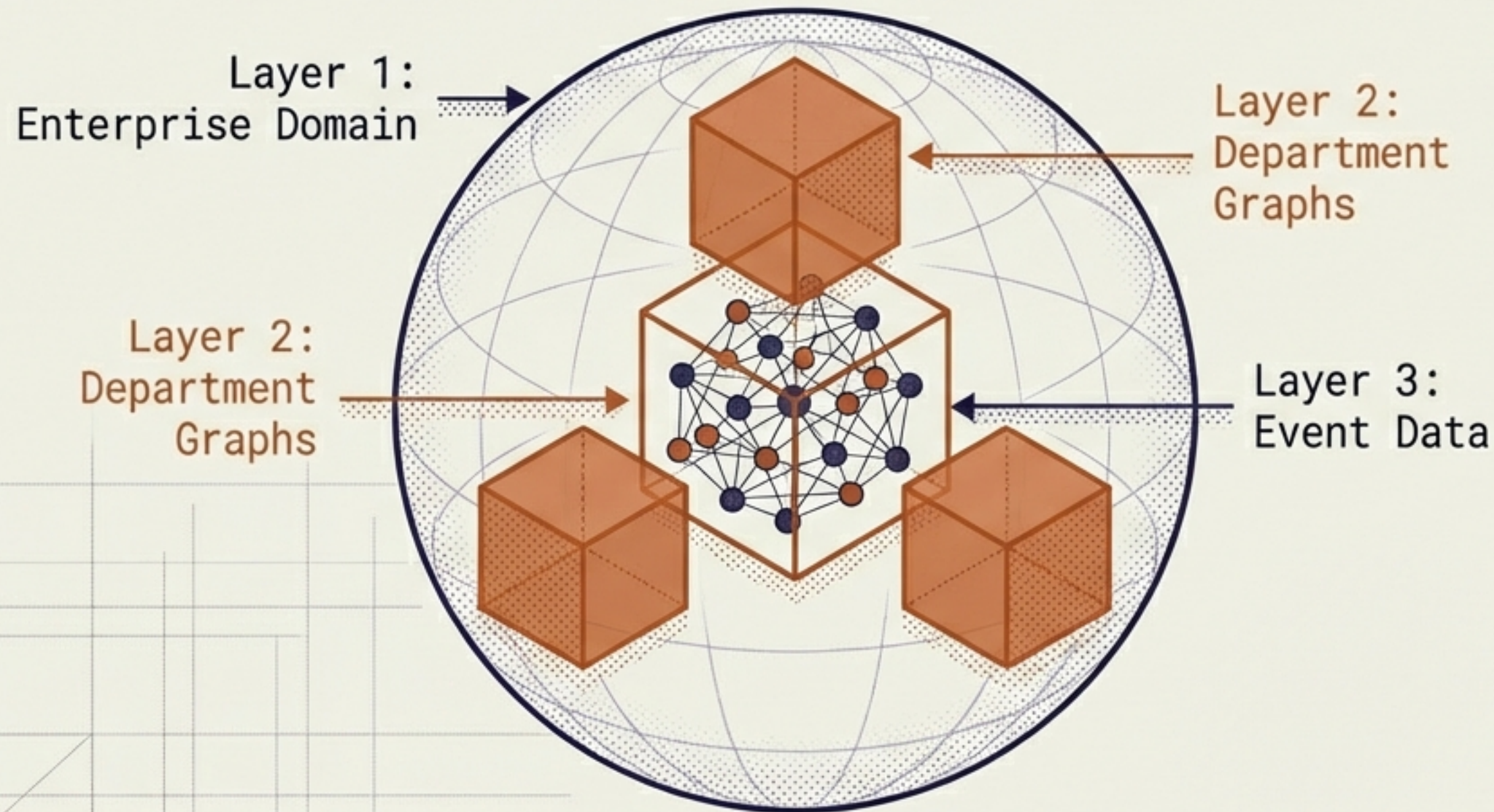
Using W3C PROV allows us to formalise the history of data into verifiable decision trails.

Meta-Provenance:

PROV-O allows for "Bundles"—encapsulating an entire provenance trail as a single entity to track the provenance of the provenance.

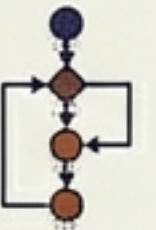
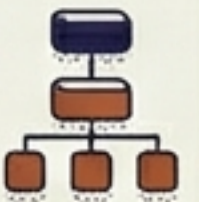
Introducing G3: The Graphs of Graphs of Graphs Architecture

A system where modular graphs (representing layers, domains, or contexts) are treated as nodes within higher-level graphs. Don't build a monolith; build linked modules.

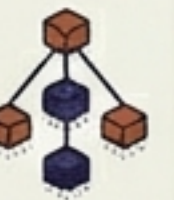


Analogy:

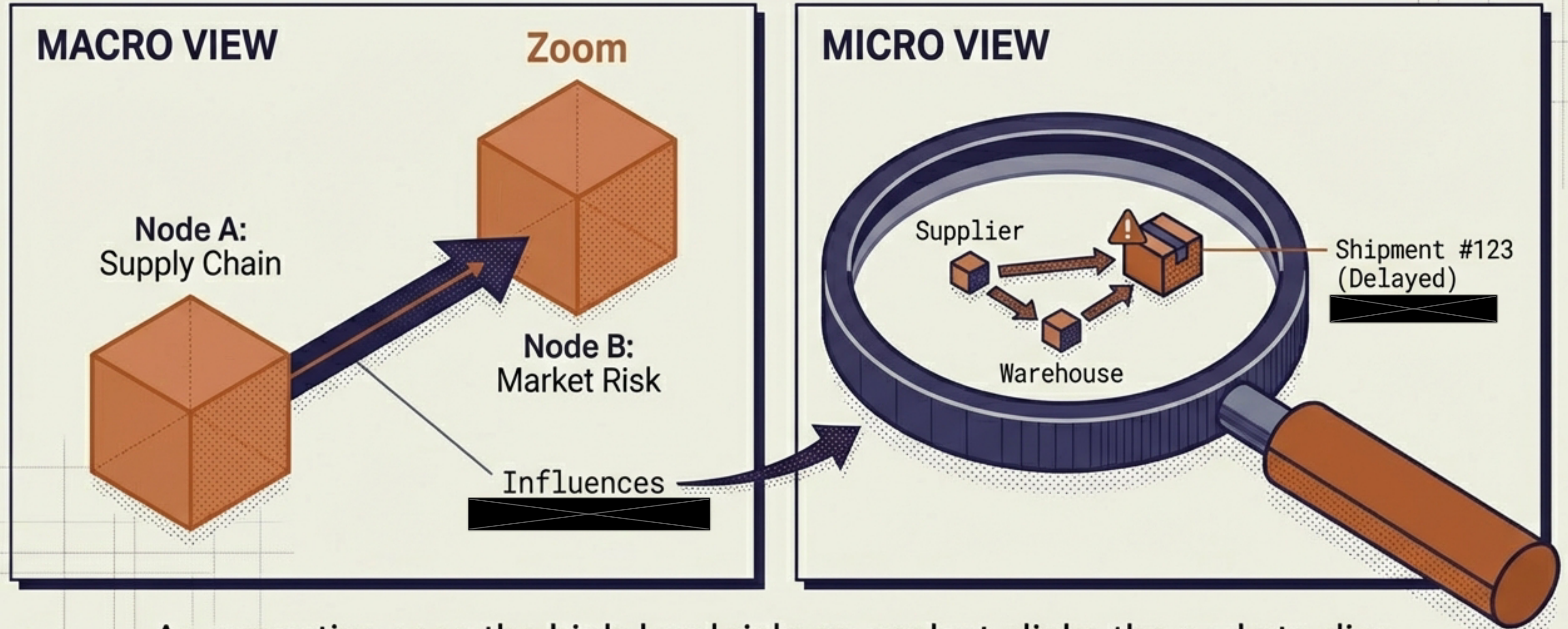
A Software Compiler uses an Abstract Syntax Tree +  Control Flow Graph + Trace Graph to understand code.



G3 applies this layering to enterprise knowledge.



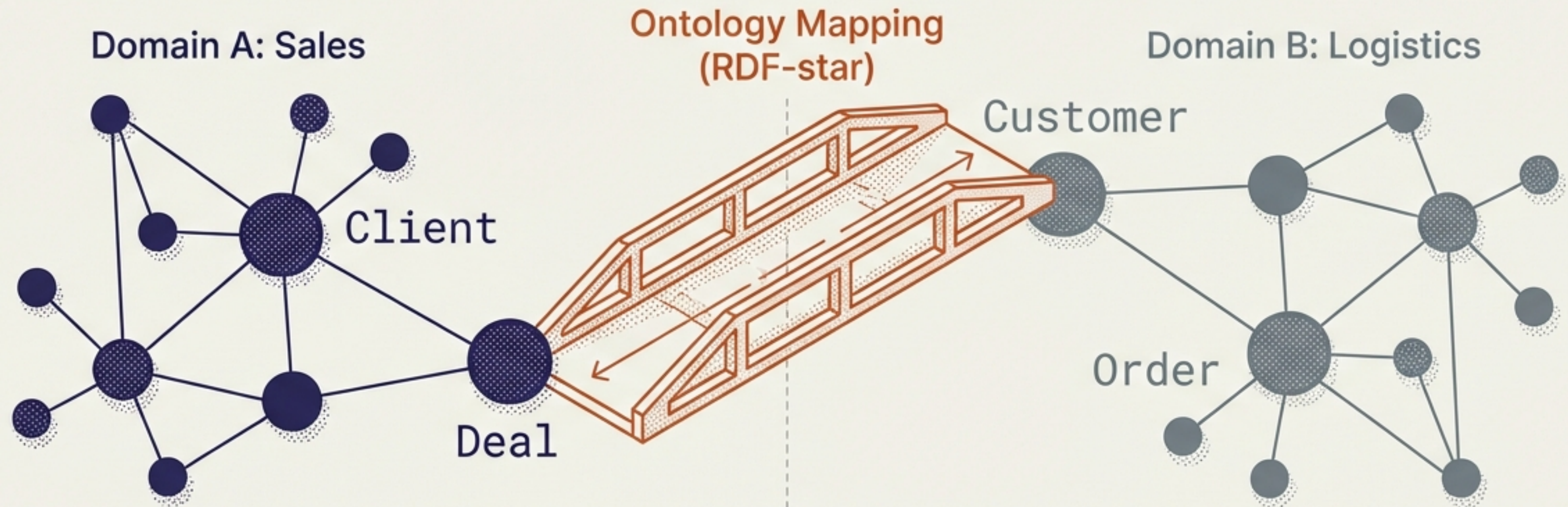
Traversing Levels of Abstraction



An executive sees the high-level risk; an analyst clicks the node to dive in and find the specific delayed shipment causing the alert.

The Semantic Glue: Ontologies of Ontologies

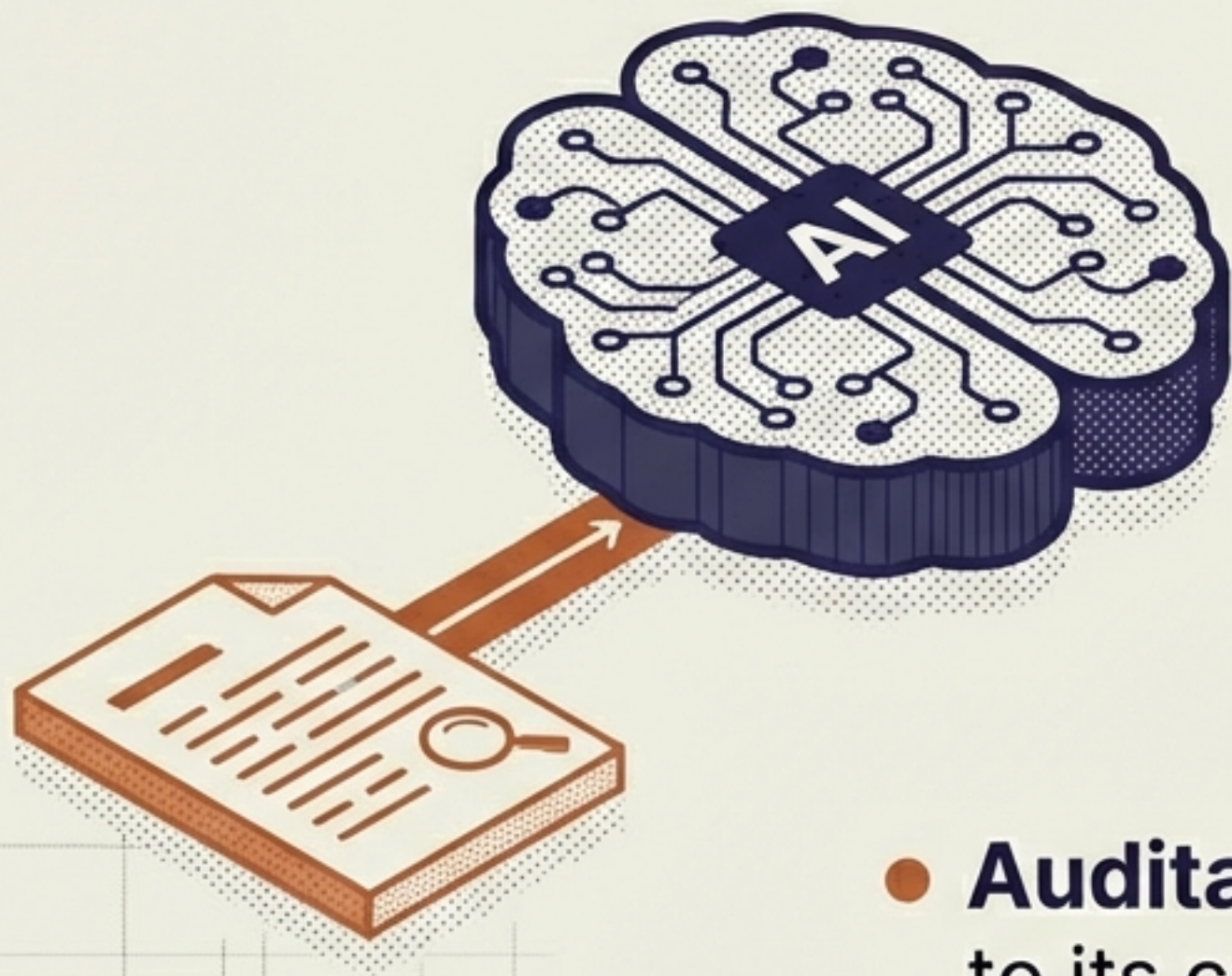
Different domains define terms differently. We do not need one “Uber Ontology.” We use bridge ontologies to map concepts across graphs.



Tech Note

Named Graphs treat sub-graphs as identifiable units.
RDF-star attaches context to the links between them.

Strategic Benefit: A System of Agency



I recommended action **Z** because factors **X** and **Y** were present and policy **Q** was met.

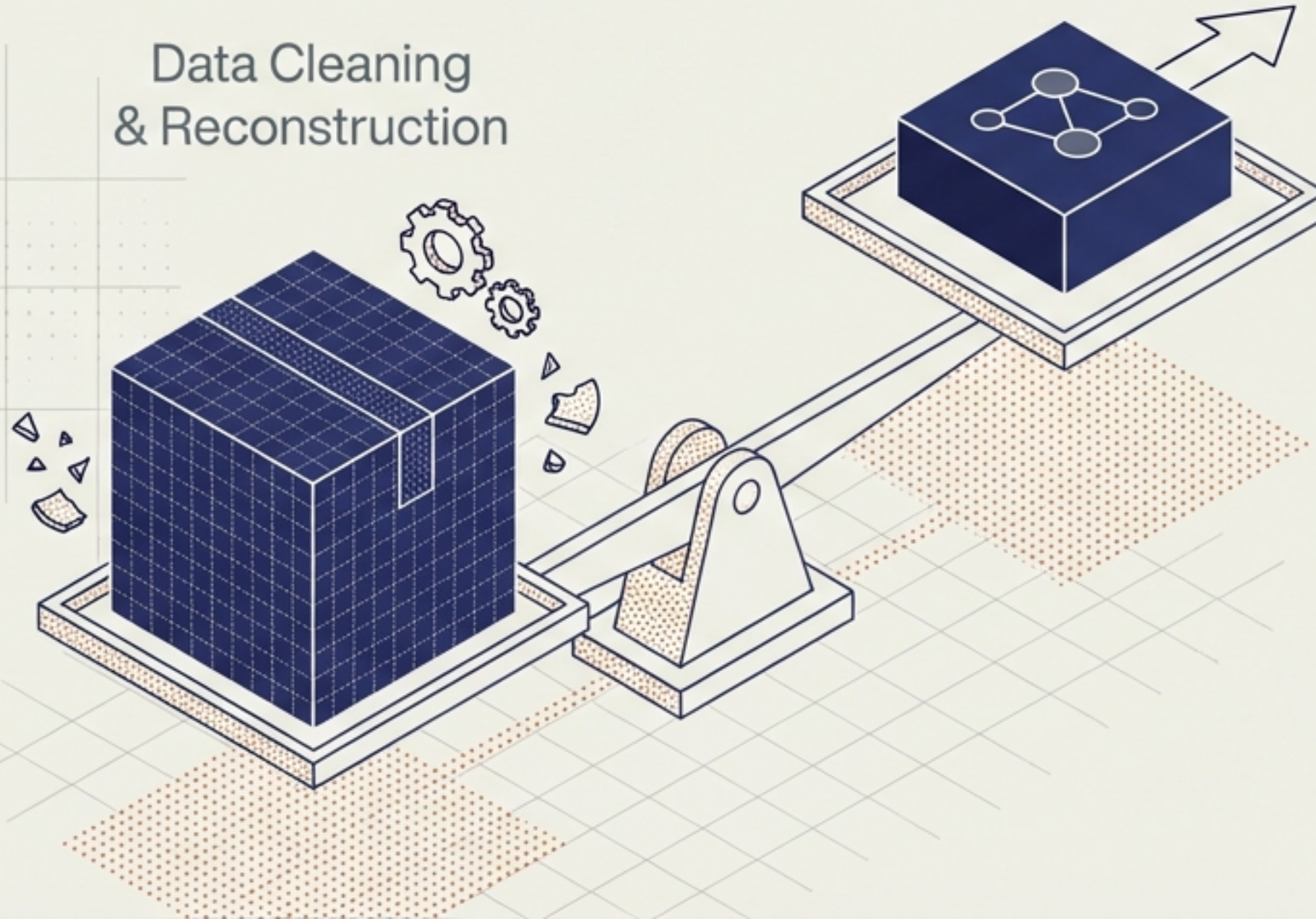
- **Auditable Truth:** Every number is a clickable link to its calculation logic.
- **Explainable AI:** Agents reason over the *logic* of past decisions, not just patterns in text.

Operational Benefit: Stop Fixing Data, Start Preventing Entropy

Better Capture (at source) > Better RAG (after the fact)

G3 Architecture

Data Cleaning
& Reconstruction



- Drastically reduces data wrangling time.
- New data sources added as modular graph modules.
- Knowledge remains machine-readable forever.

G3 in Action: The Knowledge OS

Scenario: Managing a Service Outage.



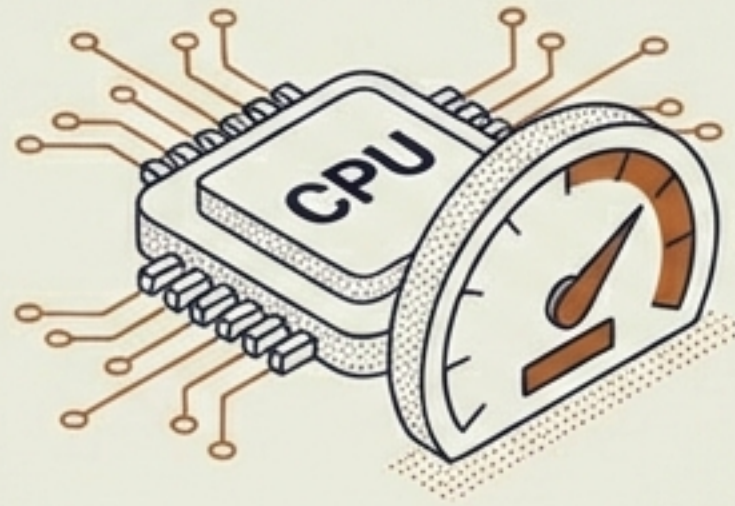
- Result: A continuous chain of meaning from sensor to boardroom.

Implementation Challenges & Reality



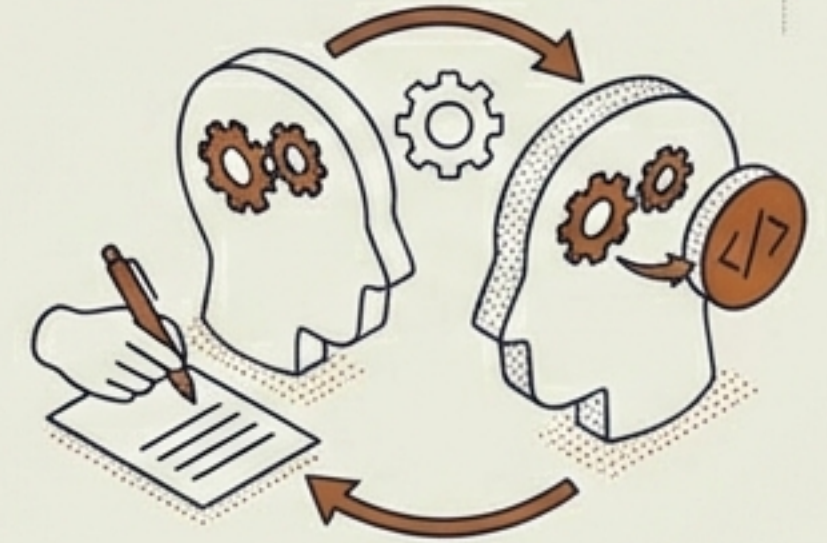
Complexity Management

Risk of information overload.
Requires excellent UX to manage 'zoom levels' so users aren't drowned in detail.



Performance

Traversing graphs-of-graphs is computationally expensive.
Requires optimised graph databases and caching strategies.

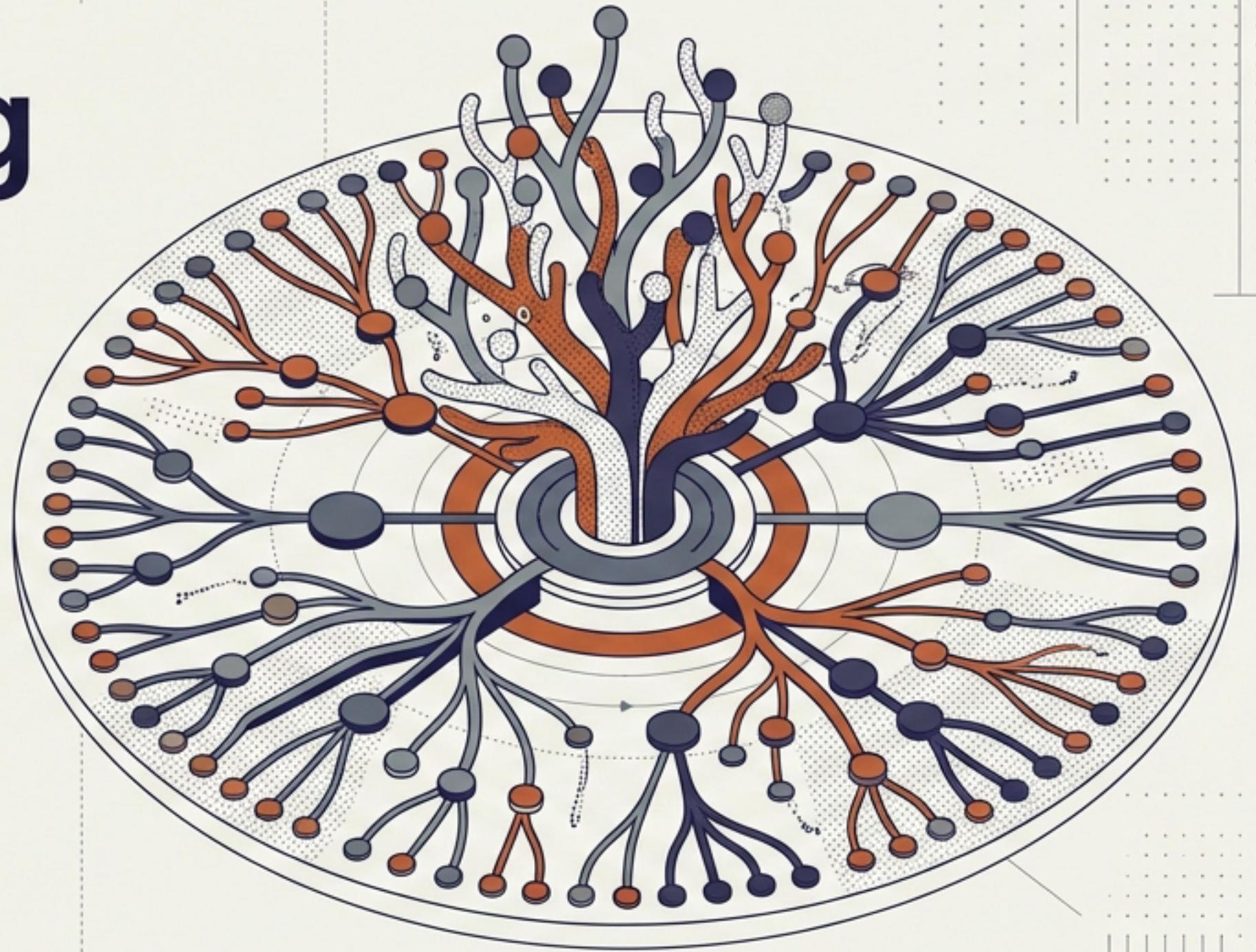


The Cultural Barrier

The hardest part. Shifting analysts from 'writing reports' to 'committing to the graph.'
Treating knowledge as code.

Towards a Living Knowledge Network

G3 moves us from black-and-white snapshots to full-colour, connected reality.



[] **Capture Context:**
Save the recipe, not just the data.

[] **Link Domains:**
Embrace modularity over monoliths.

[] **Trust the Trace:**
Make provenance a first-class citizen.